

FIELD GUIDE // PART 02 OF 03

NETWORKS

THE CITATION GRAPH OF A COURT



03

The idea: precedent leaves a trace

- The court cites its own earlier judgments — a structure that carries meaning without any text.
- One graph: **nodes** = judgments, **edges** = HD cites HD (directed).
- Similarity is inferred from *relationships*, not content: two cases are alike if the court cites with them, or cites them along with the same others.
- no text read --- the graph is the argument

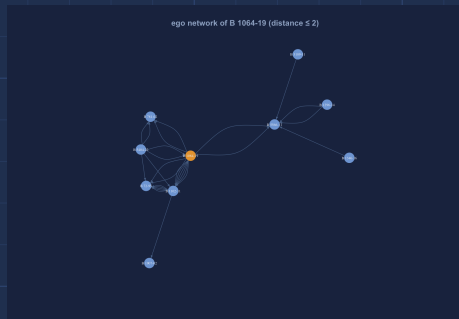
one row = one citation
citing_case -> cited_case

8,629 citation rows
5,688 in-corpus (after self-cites)
1,902 nodes, directed

66% of references resolve inside
1981-2022

Reading 1: the ego network

- **Idea:** the *closest* cases are the ones within a few citation steps of the anchor — here: distance ≤ 2 .
- Simplest possible definition: everyone inside is equally “similar” (binary), everyone outside is not.
- The anchor’s ego network is small (10 nodes): a 2019 judgment with few citations yet.
- graph methods inherit the graph's density --- sparse neighbourhood, thin answer



Reading 3: communities

- **Idea:** divide the graph into clusters that cite *within* their group more than across — no labels needed.
- Louvain (fast, standard); here: ≈ 140 **communities**, modularity 0.82.
- The anchor's community becomes a binary "similar" set.
- **Costs:** coarse — every member counts the same; *and stochastic*: the split varies run to run (anchor community from 43 to 89 cases across five runs).



The other network question: centrality

- Networks also answer: *which cases anchor the field?*
- **In-degree** — how often later cases cite this one.
- **PageRank** — weights citations by *who* cites: a case cited by central cases ranks high with few raw citations.
- Different questions, same graph — the reading is a choice.
- the demo computes both; interesting, but not the similarity question

in-degree

NJA 2011 s.	357	50
NJA 2014 s.	559	43
NJA 1994 s.	480	40

PageRank

NJA 1991 s.	444	7 cites
-------------	-----	---------

ranked #1

1,259 of 2,675 cases (47%) are never cited in-corpus

What the graph cannot say

- **Sparsity:** 47% of all cases are never cited in-corpus — the graph is silent about them, and about the anchor's newer, thinner neighbourhood.
- **A citation is not consent:** cases are cited to be distinguished and overruled — the edge exists either way.
- **One edge type:** citing—cited; the graph does not know which passage is being relied on, nor why.
- Graph readings answer *structurally similar*; next part: a reader that answers *legally similar* — and says *why*.

Live demo: the citation graph in R

- Loads `citation_edges.csv` + `cases.csv`; builds the graph with `igraph`.
- Ego neighbourhood -> Jaccard matrix -> Louvain communities.
- Three lists saved — for comparison in part 04.
- Bonus: in-degree, PageRank, the distribution plot.
- the same script is in your kit --- run it after the break

```
$ Rscript 02_network_analysis.R
[1] graph 1902 nodes, 5688 edges
[2] ego network (anchor) 10 nodes,
binary list [3] jaccard vs anchor
8 cases above 0 [4] louvain 140
communities, mod 0.82 saved:
top_10_neighbors, top_10_jaccard,
top_10_community .....
```